

Asymptotic analysis of kinetic models towards the compressible Euler–Poisson equations

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Compressible Euler equations:

$$\begin{aligned}\partial_t \rho + \nabla \cdot (\rho u) &= 0, \\ \partial_t (\rho u) + \nabla \cdot (\rho u \otimes u + p(\rho) I_n) &= -\rho \nabla \phi, \\ p(\rho) &= \rho^\gamma, \quad \gamma \in \left[1, 1 + \frac{2}{n}\right].\end{aligned}\tag{IEP}$$

Examples of ϕ :

- ▶ $-\Delta \phi = \rho$ (Poisson)
- ▶ $(-\Delta)^\alpha \phi = \rho$, $\alpha \in (0, 1)$ (Riesz)
- ▶ $-\Delta \phi = \rho - e^\phi$ (Maxwell–Boltzmann relation; ion dynamics)

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Goal: Derivation of Ionic Euler–Poisson system (IEP) from a kinetic formulation

Derivation from the bipolar Euler–Poisson system

Coupled Euler–Poisson system for ion and electron:

$$\begin{cases} \partial_t \rho_e + \operatorname{div}_x(\rho_e u_e) = 0, \\ m_e(\partial_t(\rho_e u_e) + \operatorname{div}_x(\rho_e u_e \otimes u_e)) + \nabla p_e(\rho_e) = e \rho_e \nabla \phi, \\ \partial_t \rho_i + \operatorname{div}_x(\rho_i u_i) = 0, \\ m_i(\partial_t(\rho_i u_i) + \operatorname{div}_x(\rho_i u_i \otimes u_i)) + \nabla p_i(\rho_i) = -e \rho_i \nabla \phi, \\ -\Delta \phi = 4\pi e(\rho_i - \rho_e) \end{cases}$$

- ▶ ρ_e, u_e : electron density and velocity, ρ_i, u_i : ion density and velocity,
- ▶ m_e, m_i : electron and ion mass,
- ▶ ϕ : electrostatic potential, e : electric charge

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Let η be the mass ratio:

$$\eta = \frac{m_e}{m_i},$$

and consider the zero mass limit, that is, $m_i = O(1)$ and $\eta \rightarrow 0$. Then we obtain

$$\nabla p_e(\rho_e) = e \rho_e \nabla \phi,$$

and in particular if $p_e(\rho) = \rho$, then we derive the Euler-Poisson system (IEP) for ion dynamics.

BGK(Bhatnagar-Gross-Krook) model (1954)

BGK model:

$$\partial_t f + v \cdot \nabla_x f = \frac{1}{\tau}(Mf - f),$$

where τ is the relaxation time, and it is typically inversely proportional to ρ .
Here

$$Mf = \frac{\rho}{(2\pi T)^{\frac{2}{n}}} \exp\left(-\frac{|u - v|^2}{2T}\right)$$

with

$$\rho = \int f \, dv, \quad \rho u = \int v f \, dv, \quad \text{and} \quad n\rho T = \int |v - u|^2 f \, dv.$$

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Computations of Mf :

$$\int \begin{pmatrix} 1 \\ v \\ v \otimes v \end{pmatrix} Mf \, dv = \int \begin{pmatrix} 1 \\ v \\ v \otimes v \end{pmatrix} f \, dv = \begin{pmatrix} \rho \\ \rho u \\ \rho u \otimes u + \rho T I_n \end{pmatrix}.$$

$\implies$ Conservation of mass, momentum, energy

$$\int (Mf - f) \log f \, dv \leq 0 \implies \text{Entropy inequality (H-theorem)}$$

Formal discussion: (BGK) to (IEP)

BGK-type kinetic model:

$$\partial_t f^{(\varepsilon)} + v \cdot \nabla_x f^{(\varepsilon)} - \nabla_x \phi^{(\varepsilon)} \cdot \nabla_v f^{(\varepsilon)} = \frac{1}{\varepsilon} (M f^{(\varepsilon)} - f^{(\varepsilon)}), \quad (\text{BGK})$$

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Continuity equation:

$$\partial_t \rho^{(\varepsilon)} + \nabla_x \cdot (\rho^{(\varepsilon)} u^{(\varepsilon)}) = \frac{1}{\varepsilon} \int (M f^{(\varepsilon)} - f^{(\varepsilon)}) dv$$

So, if

$$\int (M f^{(\varepsilon)} - f^{(\varepsilon)}) dv = \int M f^{(\varepsilon)} dv - \rho^{(\varepsilon)} = 0,$$

then

$$\partial_t \rho^{(\varepsilon)} + \nabla \cdot (\rho^{(\varepsilon)} u^{(\varepsilon)}) = 0.$$

Formal discussion: (BGK) to (IEP) (conti.)

Momentum equation:

$$\partial_t(\rho^{(\varepsilon)} u^{(\varepsilon)}) + \nabla_x \cdot \left(\int v \otimes v f^{(\varepsilon)} dv \right) = -\rho^{(\varepsilon)} \nabla_x \phi^{(\varepsilon)} + \frac{1}{\varepsilon} \int v (M f^{(\varepsilon)} - f^{(\varepsilon)}) dv.$$

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Note that

$$\int v \otimes v f^{(\varepsilon)} dv = \rho^{(\varepsilon)} u^{(\varepsilon)} \otimes u^{(\varepsilon)} + \underbrace{\int (v - u^{(\varepsilon)}) \otimes (v - u^{(\varepsilon)}) f^{(\varepsilon)} dv}_{=: Pf^{(\varepsilon)} \text{ (stress tensor)}}.$$

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If

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Note that

$$f^{(\varepsilon)} \sim M f^{(\varepsilon)} \quad \text{for } \varepsilon \ll 1.$$

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This shows

$$\nabla_x \cdot P f^{(\varepsilon)} \sim \nabla_x \cdot P M f^{(\varepsilon)}$$

Formal discussion: (BGK) to (IEP) (conti.)

Thus, if

$$PMf^{(\varepsilon)} = \int (v - u^{(\varepsilon)}) \otimes (v - u^{(\varepsilon)}) Mf^{(\varepsilon)} dv = (\rho^{(\varepsilon)})^\gamma I_n,$$

equivalently,

$$\int v \otimes v Mf^{(\varepsilon)} dv = \rho^{(\varepsilon)} u^{(\varepsilon)} \otimes u^{(\varepsilon)} + (\rho^{(\varepsilon)})^\gamma I_n,$$

then

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Collection of conditions on M :

$$\int \begin{pmatrix} 1 \\ v \\ v \otimes v \end{pmatrix} Mf^{(\varepsilon)} dv = \begin{pmatrix} \rho^{(\varepsilon)} \\ \rho^{(\varepsilon)} u^{(\varepsilon)} \\ \rho^{(\varepsilon)} u^{(\varepsilon)} \otimes u^{(\varepsilon)} + (\rho^{(\varepsilon)})^\gamma I_n \end{pmatrix}.$$

If $\rho^{(\varepsilon)} \rightarrow \rho$, $\rho^{(\varepsilon)} u^{(\varepsilon)} \rightarrow \rho u$, and $\phi^{(\varepsilon)} \rightarrow \phi$ as $\varepsilon \rightarrow 0$, then ρ and u satisfy (IEP).

Construction of M

Local Maxwellian, positive, indicator function closure:

$$Mf = \begin{cases} \frac{\rho}{(2\pi)^{n/2}} \exp\left(-\frac{|u-v|^2}{2}\right) & \text{for } \gamma = 1 \\ c_{\gamma,n} \left(\frac{2\gamma}{\gamma-1} \rho^{\gamma-1} - |v-u|^2 \right)_+^{d/2} & \text{for } \gamma \in \left(1, 1 + \frac{2}{n}\right) \\ \mathbf{1}_{|u-v|^n \leq c_n \rho} & \text{for } \gamma = 1 + \frac{2}{n} \end{cases}$$

with

$$d = \frac{2}{\gamma-1} - n, \quad c_n = \frac{n}{|\mathbb{S}_n|},$$

and

$$c_{\gamma,n} = \left(\frac{2\gamma}{\gamma-1} \right)^{-1/(\gamma-1)} \frac{\Gamma\left(\frac{\gamma}{\gamma-1}\right)}{\pi^{d/2} \Gamma(n/2 + 1)}.$$

Minimization principle and kinetic entropy inequality

Kinetic entropy:

$$H[f, v] = \begin{cases} \frac{|v|^2}{2} f + \frac{1}{2c^{2/d}} \frac{f^{1+2/d}}{1+2/d} & \text{for } \gamma \in \left(1, 1 + \frac{2}{n}\right) \\ \frac{|v|^2}{2} f + (+\infty) \mathbf{1}_{|f|>1} & \text{for } \gamma = 1 + \frac{2}{n} \end{cases}$$

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Minimization principle:

$$Mg = \operatorname{argmin} \left\{ \int H[g(v), v] dv : g \geq 0, \int (1, v) g(v) dv = (\rho, \rho u) \right\}.$$

Thus,

$$\int H[Mf, v] dv \leq \int H[f, v] dv$$

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Kinetic entropy inequality:

$$\begin{aligned} \frac{d}{dt} \left(\iint H[f^{(\varepsilon)}, v] dx dv + \int \left(\frac{1}{2} |\nabla_x \phi^{(\varepsilon)}|^2 + (\phi^{(\varepsilon)} - 1) e^{\phi^{(\varepsilon)}} + 1 \right) dx \right) \\ + \frac{1}{\varepsilon} \iint \left(H[f^{(\varepsilon)}, v] - H[Mf^{(\varepsilon)}, v] \right) dx dv \leq 0. \end{aligned}$$

Modulated energies

Things to do:

- ▶ existence of (weak) solutions to (BGK)
- ▶ existence and uniqueness of (strong) solutions to (IEP)
- ▶ convergence estimates

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Recall our limit system:

$$\begin{aligned}\partial_t \rho + \nabla \cdot (\rho u) &= 0, & (t, x) \in \mathbb{R}_+ \times \mathbb{T}^n, \\ \partial_t (\rho u) + \nabla \cdot (\rho u \otimes u + \rho^\gamma I_n) &= -\rho \nabla \phi.\end{aligned}\tag{IEP}$$

We rewrite the above as

$$\partial_t U + \operatorname{div}_x A(U) = F(U),$$

where

$$U := \begin{pmatrix} \rho \\ \rho u \end{pmatrix}, \quad A(U) = \begin{pmatrix} \rho u \\ \rho u \otimes u + \rho^\gamma I_n \end{pmatrix}, \quad F(U) = \begin{pmatrix} 0 \\ -\rho \nabla \phi \end{pmatrix}.$$

Define

$$\eta(U) := \frac{\rho |u|^2}{2} + h(\rho) := \frac{\rho |u|^2}{2} + \frac{1}{\gamma - 1} \rho^\gamma, \quad K(\phi, \nabla \phi) := (\phi - 1)e^\phi + 1 + \frac{1}{2} |\nabla \phi|^2.$$

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Total energy:

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Modulated energies:

$$\begin{aligned} \int \eta(\bar{U}|U) \, dx &:= \int \eta(\bar{U}) - \eta(U) - D\eta(U)(\bar{U} - U) \, dx \\ &= \int \frac{\bar{\rho}}{2} |\bar{u} - u|^2 + \frac{1}{\gamma - 1} (\bar{\rho}^\gamma - \rho^\gamma) + \frac{\gamma}{\gamma - 1} (\rho - \bar{\rho}) \rho^{\gamma-1} \, dx, \end{aligned}$$

$$\int K(\bar{\phi}, \nabla \bar{\phi} | \phi, \nabla \phi) \, dx = \int \frac{1}{2} |\nabla(\bar{\phi} - \phi)|^2 + (\bar{\phi} - 1)e^{\bar{\phi}} - (\phi - 1)e^{\phi} - \phi e^{\phi} (\bar{\phi} - \phi) \, dx.$$

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$$\int K(\bar{\phi}, \nabla \bar{\phi} | \phi, \nabla \phi) dx = \int \frac{1}{2} |\nabla(\bar{\phi} - \phi)|^2 + (\bar{\phi} - 1)e^{\bar{\phi}} - (\phi - 1)e^{\phi} - \phi e^{\phi} (\bar{\phi} - \phi) dx.$$

Modified modulated interaction energies:

$$\int \hat{K}(\bar{\phi}, \nabla \bar{\phi} | \phi, \nabla \phi) dx := \int \frac{1}{2} |\nabla(\bar{\phi} - \phi)|^2 + (\bar{\phi} - \phi)e^{\bar{\phi}} - (e^{\bar{\phi}} - e^{\phi}) dx.$$

Note that

$$(\rho_e \log \rho_e)(e^{\bar{\phi}} | e^{\phi}) = (\bar{\phi} - \phi)e^{\bar{\phi}} - (e^{\bar{\phi}} - e^{\phi}).$$

Set

$$\hat{\mathcal{E}}(\bar{U}, \bar{\phi} | U, \phi) := \eta(\bar{U}|U) + \hat{K}(\bar{\phi}, \nabla \bar{\phi} | \phi, \nabla \phi).$$

Hydrodynamic limit from (BGK) to (IEP)

Theorem: Let $T > 0$. Let f^ε be a *weak* solution satisfying the following kinetic entropy inequality

$$\mathcal{E}(f^{(\varepsilon)}, \phi^{(\varepsilon)})(t) + \frac{1}{\varepsilon} \int_0^t \iint \left(H[f^{(\varepsilon)}, v] - H[Mf^{(\varepsilon)}, v] \right) dx dv ds \leq \mathcal{E}(f_0^{(\varepsilon)}, \phi_0^{(\varepsilon)}), \quad (\text{Ent})$$

where

$$\mathcal{E}(f, \phi) := \iint H[f, v] dx dv + \int \frac{1}{2} |\nabla_x \phi|^2 + (\phi - 1)e^\phi + 1 dx,$$

and let (ρ, u) be a unique *regular* solution on the interval $[0, T]$. If the initial data satisfy the following *well-prepared* conditions:

$$\int \hat{\mathcal{E}}(U_0^{(\varepsilon)}, \phi_0^{(\varepsilon)} | U_0, \phi_0) dx, \quad \iint \left(H[f_0^{(\varepsilon)}, v] - H[Mf_0^{(\varepsilon)}, v] \right) dx dv \leq \sqrt{\varepsilon},$$

then whenever

$$\gamma \in (1, 2] \text{ when } n = 2 \quad \text{and} \quad \gamma \in \left(\frac{3}{2}, \frac{5}{3} \right] \text{ when } n = 3$$

we have

$$\rho^{(\varepsilon)} \rightarrow \rho \text{ in } L^\infty(0, T; L^p(\mathbb{T}^n)) \quad \forall p \in [1, \gamma],$$

$$\rho^{(\varepsilon)} u^{(\varepsilon)} \rightarrow \rho u \text{ in } L^\infty(0, T; L^q(\mathbb{T}^n)) \quad \forall q \in \left[1, \frac{2\gamma}{1+\gamma} \right],$$

$$\int_{\mathbb{R}^n} f^{(\varepsilon)} v \otimes v dv \rightarrow \rho u \otimes u + \rho^\gamma \mathbb{I} \text{ in } L^1([0, T] \times \mathbb{T}^n),$$

$$\phi^{(\varepsilon)} \rightarrow \phi \text{ in } L^\infty(0, T; L^r(\mathbb{T}^n)) \quad \forall r \in [1, \infty), \quad \nabla \phi^{(\varepsilon)} \rightarrow \nabla \phi \text{ in } L^\infty(0, T; L^2(\mathbb{T}^n)).$$

Modulated energy estimates

After tedious computations, we obtain

$$\begin{aligned} & \int \hat{\mathcal{E}}(U^{(\varepsilon)}, \phi^{(\varepsilon)} | U, \phi) dx \\ & \leq C\sqrt{\varepsilon} + C \int_0^t \int \eta(U^{(\varepsilon)} | U) dx ds \\ & \quad + C \int_0^t \int \left| \int (v \otimes v)(Mf^{(\varepsilon)} - f^{(\varepsilon)}) dv \right| dx ds - \int_0^t \int \Delta(\phi^{(\varepsilon)} - \phi) \nabla(\phi^{(\varepsilon)} - \phi) \cdot u dx ds \\ & \quad + C \int_0^t \int |e^{\phi^{(\varepsilon)}} - e^{\phi}| |\nabla(\phi^{(\varepsilon)} - \phi)| dx ds + C \int_0^t \int (e^{\phi^{(\varepsilon)}} - e^{\phi} - (\phi^{(\varepsilon)} - \phi)e^{\phi}) dx ds \\ & =: \sum_{i=1}^6 I_i, \end{aligned}$$

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 & \quad + C \int_0^t \int \left| \int (v \otimes v)(Mf^{(\varepsilon)} - f^{(\varepsilon)}) dv \right| dx ds - \int_0^t \int \Delta(\phi^{(\varepsilon)} - \phi) \nabla(\phi^{(\varepsilon)} - \phi) \cdot u dx ds \\
 & \quad + C \int_0^t \int |e^{\phi^{(\varepsilon)}} - e^{\phi}| |\nabla(\phi^{(\varepsilon)} - \phi)| dx ds + C \int_0^t \int (e^{\phi^{(\varepsilon)}} - e^{\phi} - (\phi^{(\varepsilon)} - \phi)e^{\phi}) dx ds \\
 & =: \sum_{i=1}^6 I_i,
 \end{aligned}$$

- Estimate of I_3 (Berthelin–Vasseur (2005)):

$$I_3 \leq C \int_0^T \int \left| \int v \otimes v (Mf^{(\varepsilon)} - f^{(\varepsilon)}) dv \right| dx dt \leq C\sqrt{\varepsilon}$$

- Estimate of I_4 : Using the integration by parts,

$$\begin{aligned}
 I_4 &= \int -\frac{1}{2} |\nabla(\phi^{(\varepsilon)} - \phi)|^2 \nabla \cdot u + \nabla(\phi^{(\varepsilon)} - \phi) \cdot \nabla u \nabla(\phi^{(\varepsilon)} - \phi) dx \\
 &\leq \frac{3}{2} \|\nabla u\|_{L^\infty} \int |\nabla(\phi^{(\varepsilon)} - \phi)|^2 dx.
 \end{aligned}$$

Estimates of electric potential

Lemma. Assume that $h \in L^\gamma(\mathbb{T}^n)$, where

$$\gamma \in \begin{cases} (1, 2] & n = 2 \\ \left(\frac{3}{2}, \frac{5}{3}\right] & n = 3. \end{cases} \quad (1)$$

Then there exist unique $\bar{\phi} \in H^1(\mathbb{T}^n)$ with zero mean and $\hat{\phi} \in H^1(\mathbb{T}^n)$ satisfying

$$-\Delta \bar{\phi} = h - \int h(y) dy, \quad -\Delta \hat{\phi} = \int h(y) dy - e^{\bar{\phi} + \hat{\phi}}.$$

Lemma. Let $f^{(\varepsilon)}$ denote a family of solutions to (BGK) satisfying the kinetic entropy inequality (Ent). Then, for γ satisfying (1), there exists a constant $C > 0$ independent of ε such that

$$\sup_{t \in [0, T]} \|\phi^{(\varepsilon)}(t, \cdot)\|_{L^\infty} \leq C.$$

Modulated energy estimates (conti.)

- Estimates of I_5 and I_6 : Setting $\tau = \phi^{(\varepsilon)} - \phi$, we can write

$$I_5 = C \int_0^t \int e^\phi |e^\tau - 1| |\nabla \tau| \, dx ds,$$

$$I_6 = C \int_0^t \int e^\phi (e^\tau - \tau - 1) \, dx ds,$$

whereas the modulated interaction energy $\hat{K}(\phi^{(\varepsilon)}, \nabla \phi^{(\varepsilon)} | \phi, \nabla \phi)$ can be written as

$$\int_0^t \int \hat{K}(\phi^{(\varepsilon)}, \nabla \phi^{(\varepsilon)} | \phi, \nabla \phi) \, dx ds = \int_0^t \int \left(\frac{1}{2} |\nabla \tau|^2 + e^\phi ((\tau - 1)e^\tau + 1) \right) \, dx ds.$$

Modulated energy estimates (conti.)

- Estimates of I_5 and I_6 : Setting $\tau = \phi^{(\varepsilon)} - \phi$, we can write

$$I_5 = C \int_0^t \int e^\phi |e^\tau - 1| |\nabla \tau| \, dx ds,$$

$$I_6 = C \int_0^t \int e^\phi (e^\tau - \tau - 1) \, dx ds,$$

whereas the modulated interaction energy $\hat{K}(\phi^{(\varepsilon)}, \nabla \phi^{(\varepsilon)} | \phi, \nabla \phi)$ can be written as

$$\int_0^t \int \hat{K}(\phi^{(\varepsilon)}, \nabla \phi^{(\varepsilon)} | \phi, \nabla \phi) \, dx ds = \int_0^t \int \left(\frac{1}{2} |\nabla \tau|^2 + e^\phi ((\tau - 1)e^\tau + 1) \right) \, dx ds.$$

Let $V, W : \mathbb{R} \rightarrow \mathbb{R}_+$ denote

$$V(\tau) := (\tau - 1)e^\tau + 1, \quad W(\tau) := e^\tau - \tau - 1.$$

Lemma. Given $A, B > 0$, there exist constants $C_A, C_B > 0$ such that

$$W(\tau) \leq C_A V(\tau) \quad \text{for all } \tau \in [-A, +\infty)$$

and

$$(e^\tau - 1)^2 \leq C_B V(\tau) \quad \text{for all } \tau \in (-\infty, B].$$

Modulated energy estimates (conti.)

Choosing $A = \|\phi\|_{L^\infty}$, we obtain

$$|e^{\phi^{(\varepsilon)}} - e^\phi| = e^\phi |e^{\phi^{(\varepsilon)} - \phi} - 1| \leq e^{A/2} e^{\phi/2} |e^{\phi^{(\varepsilon)} - \phi} - 1| \leq e^{(A+B)/2} e^{\phi/2} \sqrt{V(\phi^{(\varepsilon)} - \phi)},$$

where we have chosen $B > 2$ which satisfies $\|\phi^{(\varepsilon)} - \phi\|_{L^\infty} \leq B$; here B depends only on C_0 . Then Young's inequality yields

$$\begin{aligned} & \int_0^t \int |e^{\phi^{(\varepsilon)}} - e^\phi| |\nabla(\phi^{(\varepsilon)} - \phi)| \, dx ds \\ & \leq \int_0^t \int e^{(A+B)/2} e^{\phi/2} \sqrt{V(\phi^{(\varepsilon)} - \phi)} |\nabla(\phi - \phi^{(\varepsilon)})| \, dx ds \\ & \leq \frac{e^{(A+B)/2}}{2} \left(\int_0^t \int e^\phi V(\phi^{(\varepsilon)} - \phi) \, dx ds + \int_0^t \int |\nabla(\phi^{(\varepsilon)} - \phi)|^2 \, dx ds \right) \\ & \leq C \int_0^t \int \hat{K}(\phi^{(\varepsilon)}, \nabla \phi^{(\varepsilon)} | \phi, \nabla \phi) \, dx ds. \end{aligned}$$

Modulated energy estimates (conti.)

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For I_6 , we estimate

$$\begin{aligned} I_6 & \leq C \int_0^t \int e^\phi W(\phi^{(\varepsilon)} - \phi) \, dx ds \\ & \leq C \int_0^t \int e^\phi C_B V(\phi^{(\varepsilon)} - \phi) \, dx ds \\ & \leq C \int_0^t \int \hat{K}(\phi^{(\varepsilon)}, \nabla \phi^{(\varepsilon)}) |\phi, \nabla \phi| \, dx ds \end{aligned}$$

Modulated energy estimates (conti.)

Summing up the above estimates, we have

$$\int \hat{\mathcal{E}}(U^{(\varepsilon)}, \phi^{(\varepsilon)} | U, \phi) dx \leq C\sqrt{\varepsilon} + C \int_0^t \int \hat{\mathcal{E}}(U^{(\varepsilon)}, \phi^{(\varepsilon)} | U, \phi) dx ds$$

and thus,

$$\int \hat{\mathcal{E}}(U^{(\varepsilon)}, \phi^{(\varepsilon)} | U, \phi) dx \leq C\sqrt{\varepsilon}.$$

From the above, we derive the following convergence estimates:

$$\rho^{(\varepsilon)} \rightarrow \rho \text{ in } L^\infty(0, T; L^p(\mathbb{T}^n)) \quad \forall p \in [1, \gamma],$$

$$\rho^{(\varepsilon)} u^{(\varepsilon)} \rightarrow \rho u \text{ in } L^\infty(0, T; L^q(\mathbb{T}^n)) \quad \forall q \in \left[1, \frac{2\gamma}{1+\gamma}\right],$$

$$\int_{\mathbb{R}^n} f^{(\varepsilon)} v \otimes v dv \rightarrow \rho u \otimes u + \rho^\gamma I_n \text{ in } L^1([0, T] \times \mathbb{T}^n),$$

$$\phi^{(\varepsilon)} \rightarrow \phi \text{ in } L^\infty(0, T; L^r(\mathbb{T}^n)) \quad \forall r \in [1, \infty),$$

$$\nabla \phi^{(\varepsilon)} \rightarrow \nabla \phi \text{ in } L^\infty(0, T; L^2(\mathbb{T}^n)).$$

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$$\gamma \in \begin{cases} (1, 2] & n = 2 \\ \left(\frac{3}{2}, \frac{5}{3}\right] & n = 3 \end{cases} \implies \gamma \in \{0\} \cup \left[1, 1 + \frac{2}{n}\right].$$

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Thank you for your attention.